

TO: ME 2171 Instrumentation Lab Instructional Staff January 13th, 2026

FROM: Mateo Jaime Tecum Jacinto, Section 07- Wednesday

SUBJECT: Lab 1, Introduction to LabVIEW

1.0 Statement of Purpose

The purpose of this lab was to get familiar with the program LabVIEW by creating a functional calculator and tank simulator. These two projects serve as a fine method to accustom one to LabVIEW, requiring varying functions within LabVIEW's control and function palette. The simulation program generates a random number ranging from 0-1, which then accumulates in a tank, drains, and a mean of the values is calculated after the runs are complete. The calculator program allowed the user to input two numbers, chose a math operation, which the calculator would then solve and project the answer on the front panel.

2.0 Background

LabVIEW serves as a software program that can control and manipulate data with its extensive tools available. Some of LabVIEW's tools include temperature indicator, strain gauge, and pressure sensor, all of which can be used to acquire data for later use in labs or real industry purposes. The other tools LabVIEW has allows for calculations and displays results for multiple simulations, acting as a method for gathering and logging data. The Virtual Instruments(which is the name of LabVIEW's tools) includes the front panel and the block diagram. The front panel allows user interface with images of the physical instrument being used, like a switch or graph. The block diagram represents the code the program has, which connects icons together by wires, representing the direction the data flows in the Virtual instrument. LabVIEW's extensive toolbox allows for many simulations and programs to be created that can help in testing systems or acquiring data for the main part, but can also be used for research, especially with making testing systems that are complex easier to modify in the block code.

3.0 Discussion

3.1 Random Number Example.vi

3.1.1 Block Diagram

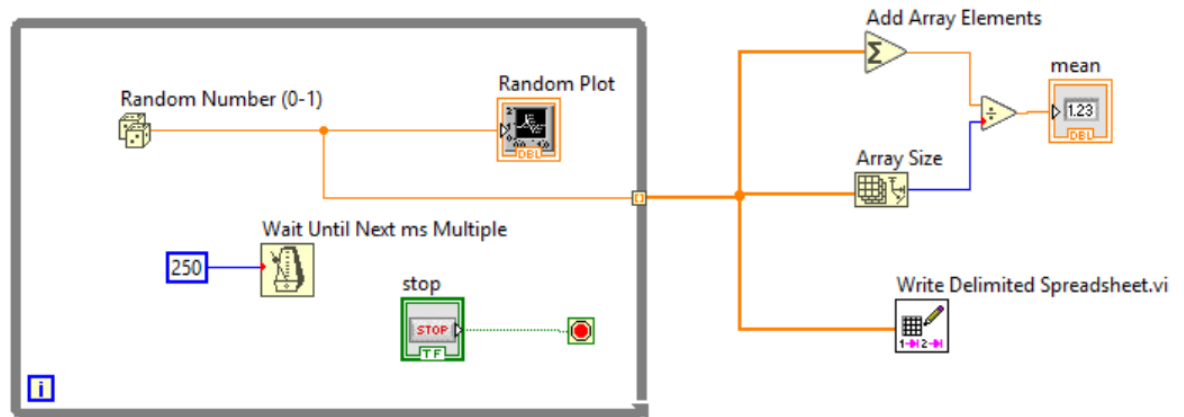


Figure 1. Random Number Example vi block diagram

The Random Number Example's block diagram, as seen in *Figure 1.*, utilizes a while loop to generate multiple random numbers, with the specific value (outside of the loop) can be recorded as a number in an array, and is then used to calculate the mean of all the numbers generated once the loop has stopped from the user interface (the stop button/boolean). The general loop includes creating a number varying from zero to one, plotting the number on a waveform chart. This chart and data is generated at a rate of 250 ms. The wires representing the data flow, can be seen as first generating the random number, recording the value on the plot, and then the value leaves the while loop, which then is recorded in an array. The array values are then all summed up and divided by the array size which is then used to calculate the average of all the numbers generated during the while loop. The Write Delimited Spreadsheet tool is used to record all the 1D data into a text file. This essentially has all the data recorded onto a spreadsheet, which can be saved at the end of the run.

For this block diagram, the use of three types of wiretypes was used. This included doubles, Booleans, and constants. In figure 1, the color of the wiring is used to distinguish each wiretype, with the majority of this block diagram being orange to represent a double. For the Wait Unit Next ms Multiple and the wiring from Array size to the divide function, a blue color is used to represent a constant number. Lastly, the stop function has a green wire color to convey the wire is dealing with a Boolean of either TRUE or FALSE.

3.1.2 Front Panel

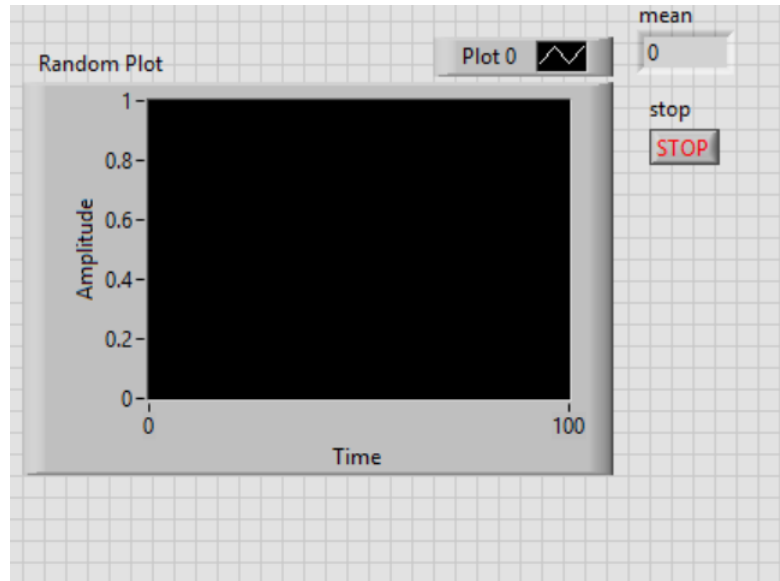


Figure 2. Random Number Example vi, Front Panel

The front panel of this program has arranged all the properties that can be controlled through the Random Number Example, which includes the stop button, the waveform chart (which is now displaying all the data generated), and conveys the mean at the end of the executed while loop.

3.2 Tank Simulator

3.2.1 General Description

The Tank Simulator works by simulating a tank that is filled up by values generated in the block diagram. The values, similar to the aforementioned numbers generated in the Random Number Example, are generated from 0-1, however, include various toggle switches that can cause the tank level to decrease or “drain” during the simulation. On the front panel, the user can flip to either automatic or manual tank level control using one toggle switch, and another toggle switch is used to represent the tank draining valve. Additionally, a Drain Valve Status and tank is displayed to show the current tank level and the status of the tank (whether the tank is overfilling or if the tank is ok).

3.2.2 vi

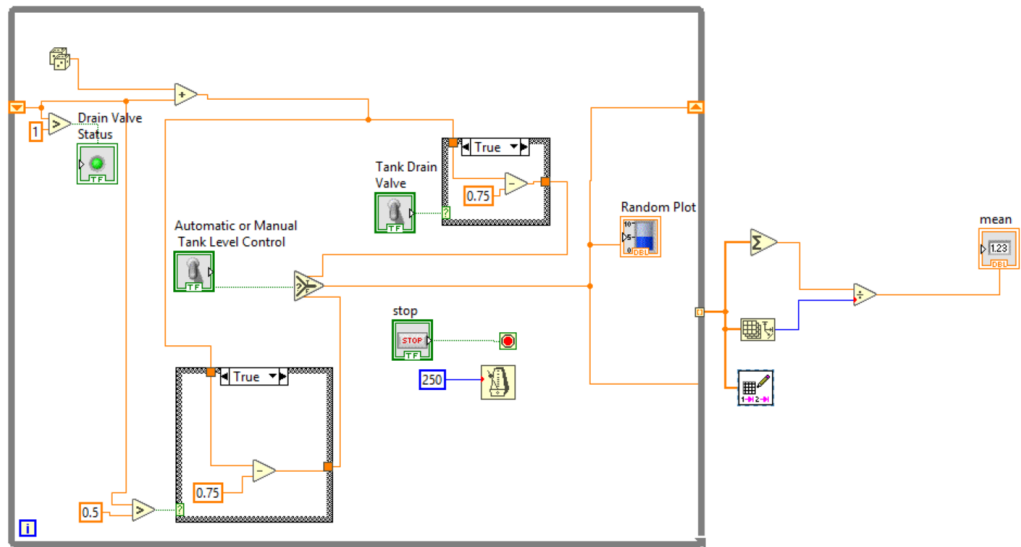


Figure 3. Tank Simulator, block diagram

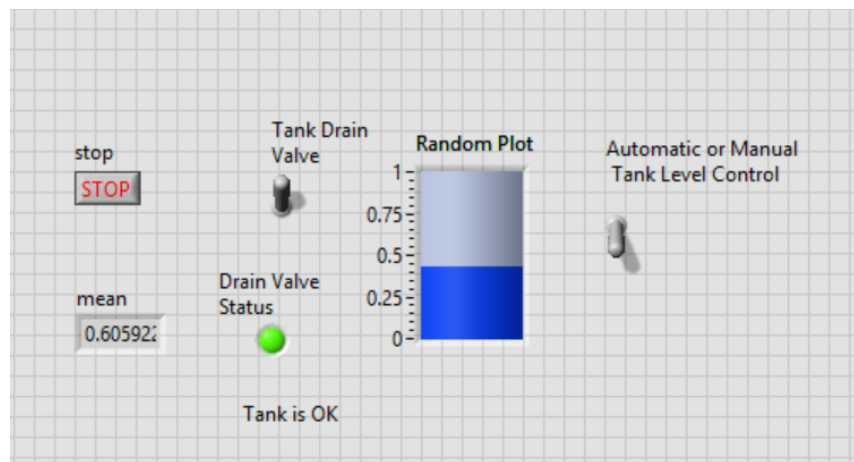


Figure 4. Tank Simulator, front panel

The Tank Simulator's Virtual Instrument (VI) shows a fluid tank level changing, which can be controlled by a manual toggle switch or automatic control (which is in play whenever the second toggle switch is set to automatic). Shown on Figure 4, the user can interact with the two toggle switches for control, a stop button to end the entire simulation. Additionally, the front panel calculates the mean of the tank simulator's values after the simulation, displays the tank's current level, and shows the status of the drain valve.

Taking a closer look in the Tank Simulator, Figure 3 operates with various tools wired together to manipulate the data as it is being generated. The logic can be followed by looking at these wire connections and what components fall within specific boxes in the block diagram (in this case, either the while loop, or a case structure). The core function of this block diagram includes the While Loop, which encapsulates the generator of a number between 0-1. The while loop begins with a wire connected at the left side of the loop, which would have the value of any previous loop. The wire then flows into a comparison which verifies if the value is greater than 1. If it is indeed exceeding one, then the Drain Valve status is turned on (which would mean that the comparison returned a true boolean). It is also compared to 0.5, which if true, would make the case structure true (the pathway of true or false for the case structure is later mentioned). The value, which begins at 0 in the first iteration of the while loop, is added with a number generated from 0-1. After this, the value runs through two cases, which can either be true or false. If true, the value is subtracted by a constant of 0.75 and the result is then going to a comparison once again. It is important to note that the second case structure is manually controlled by the user interface to be true or false. The next comparison is another manually controlled switch on the front panel, that decides which of the two case structure values will be used. If the switch is set to true, this would indicate a manual tank level control, which would take the true value. If not, then the other case structure's value would be used. This value is then recorded on the displayed tank on the front panel, while also being recorded on the shift register for use in the next iteration. The last function inside the while loop is a 250ms delay for each iteration, resulting in each iteration being 250ms apart from each other.

The value is also recorded to an array that accumulates all tank level values. This is recorded once the while loop has ended, which can occur once the stop function on the front panel is clicked.

For the functions outside the tank simulator, the array size and elements are used to calculate the average (mean). The values of all the iteration's values are also recorded onto a spreadsheet that can be saved.

4.0 Conclusions

The Random Number and Tank simulator Virtual Instruments have both been simulated and run correctly.

5.0 Quiz

Name: Mateo Tecum

Date: 1/13/26

Writing Guide Quiz

1. Good technical writing is orderly, and predictable.
2. Well written technical documents follow a well-established order called what?
 - a. Theme
 - ☒ b. Format
 - c. Convention
 - d. Form
3. Are engineering reports personal or impersonal?
 - a. Personal
 - ☒ b. Impersonal
4. Circle the preferred statement.
 - a. We used Equation 4 to determine the Reynolds number.
 - ☒ b. Equation 4 was used to determine the Reynolds number.
5. Which of the following is the better statement?
 - a. The data agrees reasonably well with the model.
 - ☒ b. The data agreed with the model within +/- 10%.
6. Is the following sentence active or passive? "The pressure increases as the temperature rises."
 - ☒ a. Active
 - b. Passive
7. Is the following sentence active or passive? "The air pressure, temperature, and humidity were recorded."
 - a. Active
 - ☒ b. Passive
8. Is technical writing passive or active?
 - a. Wholly passive
 - b. Wholly active
 - ☒ c. A combination of both
9. The number of decimal places of a measured or calculated value is determined by its number of significant digits.